

Reference-gauge Figure-3 same-tree resolution investigation

AthenaK Z4c / native vertex-centered Cartoon SO(2)

28 August 2026

Verdict: Aurora PVC qualified; exact replay verified; strong resolution improvement; early O4-compatible behavior; first peak remains constraint invalid; full convergence is not established.

1 Question and controlled experiment

The experiment asks whether simply increasing the cells per physical MeshBlock cures the constraint-invalid Brill-collapse Figure-3 result. N128, N256, and N512 use the same root MeshBlock layout, physical MeshBlock bounds, accepted LogicalLocation hierarchy, and AMR event times. Only the cells per MeshBlock change in the sequence 16^2 , 32^2 , and 64^2 .

All cases use the shock-avoiding Bona–Masso lapse with $\kappa = 1$, unit initial lapse, prescribed zero shift, disabled telegraph lapse, native vertex-centered Cartoon SO(2), O4 finite differences, q6 prolongation, RK4, CFL 0.15, KO dissipation 0.50, zero Z4c constraint damping, and outer boundary $128M$. No gauge, transfer, KO, CFL, or AMR parameter was tuned.

The N256 authority has SHA-256 7055de601e6181e5ad7e1432b5c20a111b0ba67e0e8d5377c170ea80e7bedcde. N128 and N512 accepted both replay events exactly, ending at 212 leaf MeshBlocks and physical level 5.

2 Execution and provenance

The qualified executable SHA-256 is aae7ccb8739fb4951221ad7be69ea0e220548b52d402086f57d7857fa2c97a13 from source commit f8303c6be7eb214fa1e91b646123ee0d434b3698. Aurora qualification job 8789659 passed eleven focused PVC tests.

Case	final coordinate time	final proper time	disposition
N128	45.0000	19.33240	reached tlim; constraint invalid
N256	30.0000	11.28631	reached tlim; constraint invalid
N512	38.6523	14.98253	gate exceeded; constraint invalid at peak

N512 used jobs 8789895, 8789956, 8790025, 8790135, 8790202, and 8790242. N128 used jobs 8790272 and 8790338. No science job remains active. Segment N512/000 completed Athena cleanly, but PBS killed the allocation during a redundant second full artifact verification pass. Its science restart and disposition were already written. Every later segment has a complete sealed manifest and scheduler exit zero.

3 Direct Figure-3 comparison

No time, amplitude, or vertical transform was fitted.

Case	peak τ	peak $\log_{10} \mathcal{K} $	peak $\int C^2 dV$
N128	10.31396	4.29765	107.608
N256	10.30333	5.01349	48.2330
N512	10.30811	5.38112	4.09930
Published	10.30683–10.31384	5.47778–5.48688	–

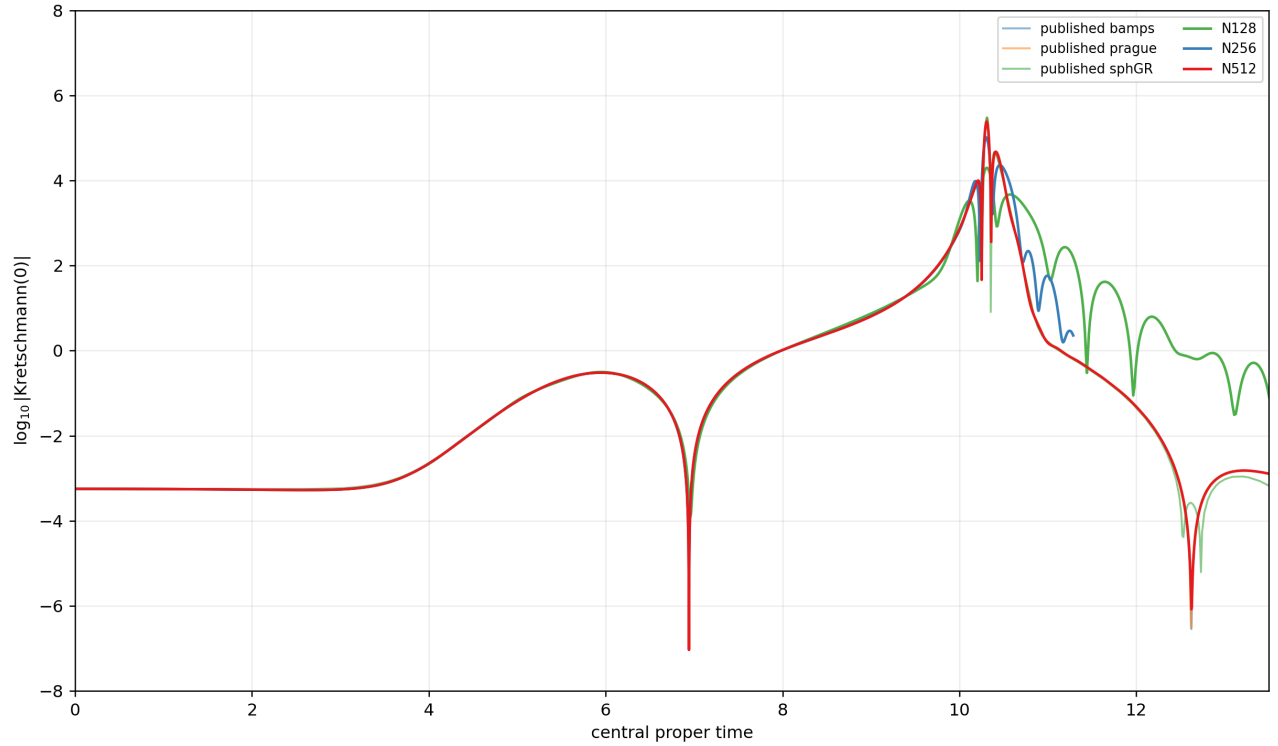


Figure 1: Direct central Kretschmann comparison with the published curves.

N512 resolves the deep minimum at $\tau = 12.62280$, $\log_{10} |\mathcal{K}| = -6.07875$, and the rebound at $\tau = 13.21629$, $\log_{10} |\mathcal{K}| = -2.81849$. These lie inside the ranges of the three published curves. N128's nominal late extrema are constraint dominated, and N256 does not reach the interval.

4 Constraint qualification

The history measure is already the physical axisymmetric ring measure,

$$2\pi\rho\sqrt{|\det\gamma|}d\rho dz,$$

with canonical vertex ownership and trapezoid endpoint weights. The result is not a fictitious collapsed- y normalization artifact.

Squared integral	N128 maximum	N256 maximum	N512 maximum
C	107.608	48.2330	4.09930
H	88.4516	41.1314	3.63079
M	25.3663	9.53227	0.682824
Z	0.643210	0.0550105	0.00462028

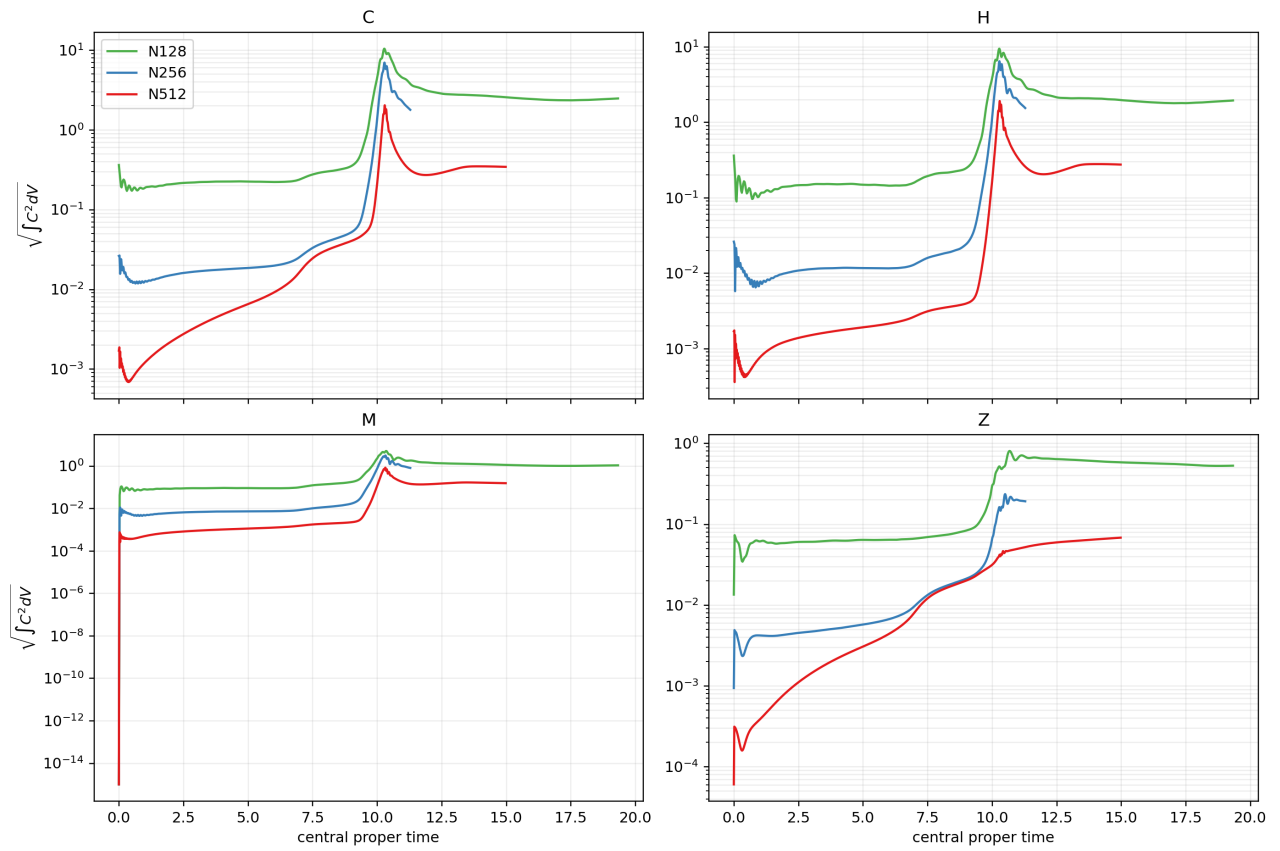


Figure 2: Physical-volume constraint amplitudes, plotted as the square root of the stored squared integrals.

At N512, aggregate C first crosses 0.01, 0.1, and 1 at proper times 9.88652, 10.05721, and 10.19025. C and H peak at $\tau = 10.29711$, essentially coincident with the curvature peak. The improved Figure-3 curve is therefore still constraint invalid.

The N256 and N512 C/H maxima are on the symmetry axis and far from a coarse-fine interface. M is within one radial spacing of the axis. This is a geometric classification, not proof of an axis source defect.

5 rho near 5 high-frequency diagnostic

All 25 evolved fields were stitched over the finest physical patch $4 \leq \rho \leq 6$, $-2 \leq z \leq 2$ before differencing and Fourier analysis. Shared vertices agree to roundoff. Matched slices were analyzed at proper times approximately 8.03, 9.50, 10.27, and 11.00.

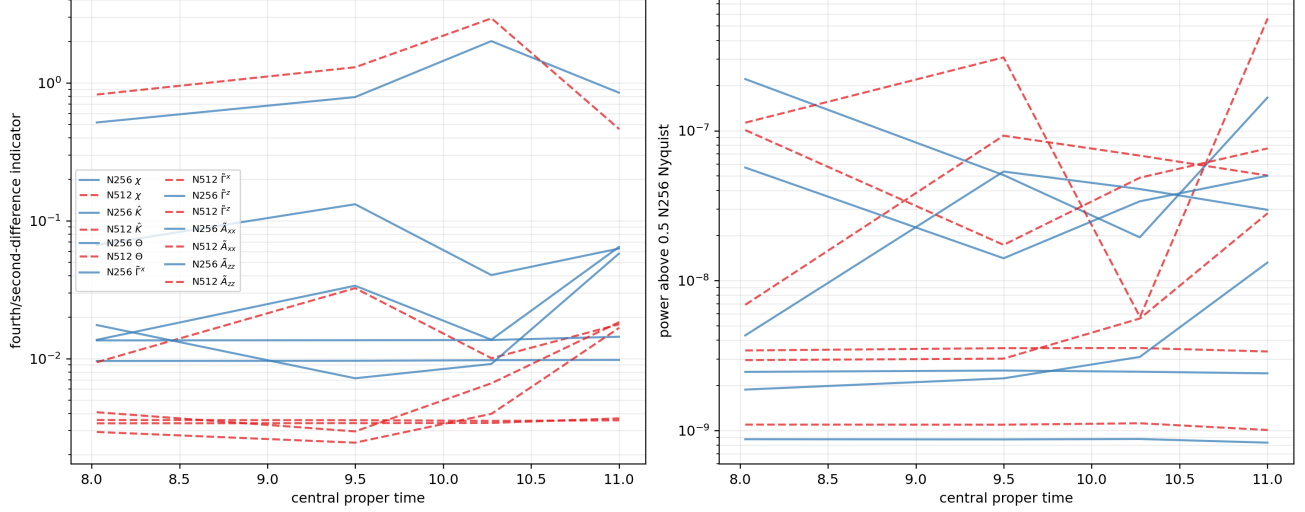


Figure 3: Fourth-difference and above-N256-Nyquist diagnostics on the stitched rho=4–6 patch.

At the near-peak slice, N512/N256 fourth-difference ratios are about 0.36 for χ , 0.25 for $\hat{K}$ and sampled conformal connections, and 0.23–0.48 for representative $\tilde{A}_{ij}$ components. Θ has the largest normalized pre-peak indicator, but its absolute fluctuation is only 3.5–6.2 percent of N256 and its power above half the N256 Nyquist stays below 3.1×10^{-7} . By proper time 11 its normalized indicator is smaller as well. No measured rho near 5 high-k branch grows with resolution.

6 Three-resolution convergence assessment

At common central proper time, the Richardson estimator is

$$p = \log_2 \left(\frac{|u_{128} - u_{256}|}{|u_{256} - u_{512}|} \right).$$

Proper-time window	central Kretschmann	central lapse
0–8	4.86	3.93
8–10	3.34	3.36
10–11.286	2.10	1.40

The central fields are O4-compatible early, then lose order through collapse. Constraint-amplitude pair orders are positive but inconsistent. For aggregate C, N128/N256 versus N256/N512 median orders are 3.46 versus 1.06 in $\tau < 8$ and 1.03 versus 2.38 in $10 < \tau < 11.286$. No single order describes both pairs.

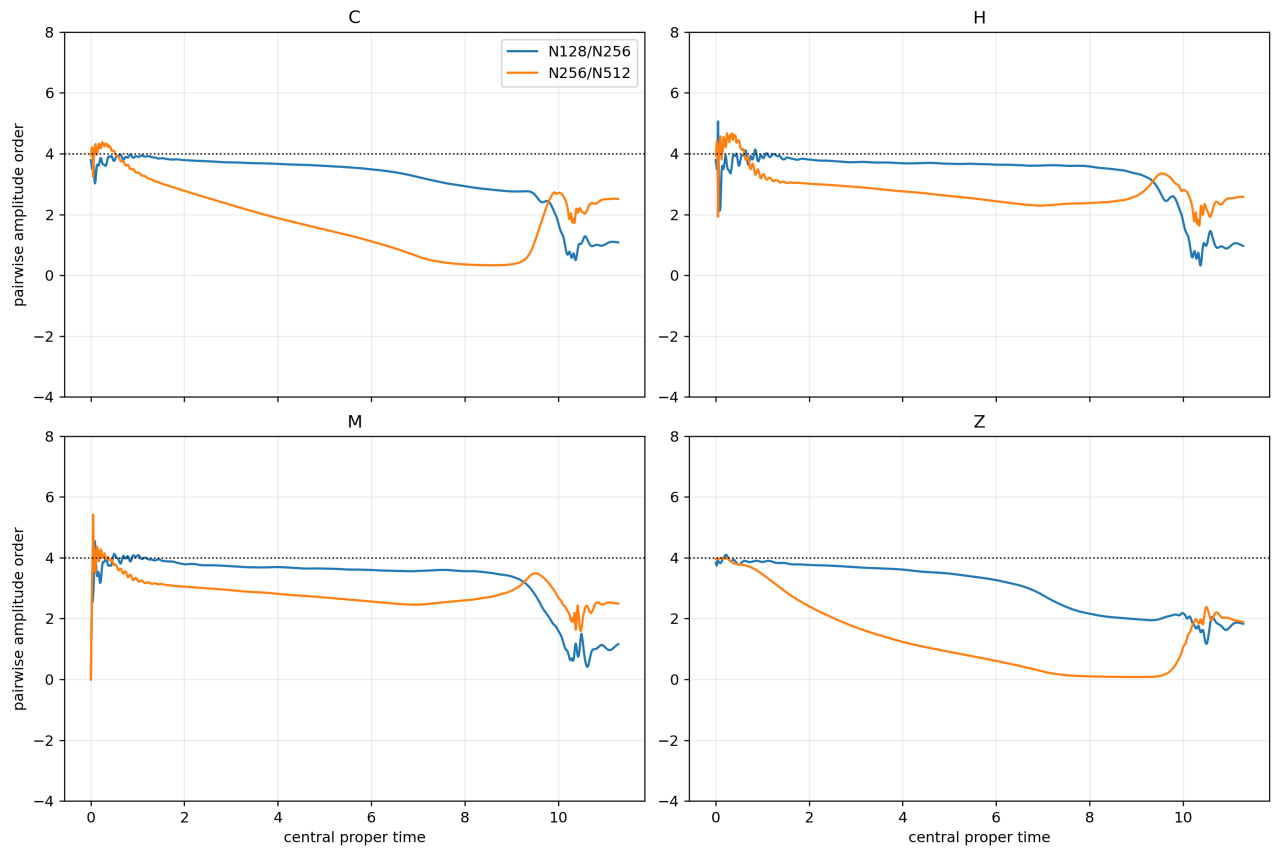


Figure 4: Pairwise convergence orders of the physical-volume constraint amplitudes. The mismatch between pairs prevents a uniform order claim.

N256 ends at $\tau = 11.286$, so three-level order cannot be measured through the deep-minimum/rebound interval.

7 Interpretation and next step

The identical interface geometry becomes markedly better behaved as cells per MeshBlock increase. N512 moves the peak toward the published amplitude, reduces peak constraints by about an order of magnitude relative to N256, suppresses rho near 5 numerical content, and survives through the full Figure-3 interval. Bulk/parent under-resolution is therefore a major driver of the N256 failure and is not ruled out.

This does not prove that AMR interfaces or the axis are harmless. N512 still has an order-unity constraint episode at the peak, and the global maxima are axis-adjacent. A residual interface or axis contribution can coexist with bulk under-resolution.

The natural next step is a bounded shared-RHS audit at the retained N512 peak state, separating active-axis stencils, same-level seams, and coarse-fine ghost and interface contributions. Gauge, KO, transfer, positivity checks, and the exact tree should remain unchanged.

8 Claim boundary

Supported: exact N128/N512 hierarchy replay; strong resolution improvement; early O4-compatible central fields; N512 coverage of the published peak, minimum, and rebound; smaller/later constraints and weaker rho near 5 high-frequency content.

Not supported: a constraint-qualified Figure-3 reproduction; uniform convergence through collapse; exclusion of all axis/interface mechanisms; a unique source-level defect; or a production numerical correction.